Algolog Limited

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Court474

AI-Powered Short-Let Property Management & Guest Distribution System

Solution Brief for Court 474 Ltd

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1. THE BUSINESS

Court 474 Ltd operates a short-let (Airbnb-style) property consisting of:

- **25 x 1-bedroom apartments** — owned by different subscribers/property owners
- **4 x 2-bedroom apartments** — owned by different subscribers/property owners
- **1 x 40-seater conference room** — meetings, seminars, retreats, trainings

Court 474 manages the facility on behalf of the individual apartment owners across two layers:

Layer 1 — General Facility Management (all units)

- Security, water supply, cleaning of external premises and common areas
- Charged at a pre-agreed fixed service charge to all apartment owners

Layer 2 — Individual Apartment Management (opt-in)

- Short-letting, guest management, cleaning of individual apartments
- Charged at an agreed fee per apartment
- This is the revenue-generating layer where technology has the most impact

2. THE PROBLEMS TO SOLVE

A. Fair Guest Distribution

With 29 apartments owned by different people, how do you ensure bookings are distributed evenly? Without a system:

- Some apartments get overbooked while others sit empty
- Owners complain about unequal treatment
- Manual allocation is time-consuming and biased (consciously or not)
- No transparency for owners to verify fair distribution

B. Revenue Splitting

Each booking generates revenue that needs to be split between the apartment owner (their share) and Court 474 (management fee). This needs to happen automatically, transparently, and with a full audit trail.

C. Operations Cost

Running 29 apartments + a conference room with traditional staffing is expensive:

- Receptionist/front desk for check-in/out
- Manual booking tracking (WhatsApp, spreadsheets, calls)
- Manual cleaning scheduling and procurement tracking
- Manual invoicing and payment reconciliation

D. Conference Room Management

The 40-seater conference room is a separate revenue stream that needs its own booking system (hourly/daily), availability calendar, payment processing, and conflict prevention.

3. THE SOLUTION

A purpose-built platform for Court 474's multi-owner short-let model. It manages the entire operation: bookings, AI guest distribution, owner settlements, conference room, and facility operations — all from one system.

Module 1: AI Guest Queuing & Distribution Engine

The system maintains a **fairness score** for each apartment. When a new booking request comes in, the AI engine allocates it based on:

1. **Occupancy balance** — Which apartments have had the fewest bookings this period? Those get priority.
2. **Revenue balance** — Which owners have earned the least? The system balances revenue, not just bookings (since 2-beds earn more than 1-beds).
3. **Apartment type match** — Guest requests a 1-bed → only 1-bed apartments are considered. 2-bed → only 2-beds.
4. **Availability** — Only apartments that are actually available for the requested dates.
5. **Cleaning status** — Only apartments that are cleaned and ready.
6. **Owner preferences** — Some owners may block certain dates (personal use, maintenance).

Every owner gets a fair share of bookings. The system can show each owner exactly how many bookings they've received vs. the average, and why a particular apartment was chosen for a particular guest.

Distribution modes:

- **Round-robin** — Strict rotation (Apartment 1, then 2, then 3...)
- **Weighted fair share** — Balances by revenue earned, not just count
- **Priority override** — Admin can manually assign a specific apartment when needed (with audit log)

Module 2: Owner Dashboard & Revenue Distribution

Each property owner gets a portal showing:

- **Live occupancy status** of their apartment(s)
- **Booking history** — who stayed, when, how long, how much
- **Revenue earned** — gross booking amount, management fee deducted, net owner earnings
- **Automatic settlement reports** — weekly/monthly statements
- **Fairness transparency** — their booking count vs. property average, revenue vs. average
- **Calendar blocking** — owners can block dates for personal use
- **Maintenance requests** — flag issues in their apartment

Revenue split flow:

1. Guest pays for booking → money lands in Court 474 account
2. System calculates: Gross amount → Management fee (%) → Owner net

3. Settlement report generated automatically
4. Court 474 disburses to owner on agreed schedule (weekly/monthly)
5. Full audit trail — every naira accounted for

Module 3: Guest Booking & Management

For guests:

- Online booking portal — browse available apartments, see photos, select dates, pay online
- Instant booking confirmation via email/SMS
- Smart lock access code — generated automatically, sent to guest, expires at checkout
- QR check-in — scan on arrival, validated against active booking
- No front desk needed for standard check-ins

For Court 474 staff:

- Dashboard showing all active bookings, upcoming arrivals, departures
- Guest communication tools and issue/complaint tracking
- Automated reminders (checkout time, extension offers)

Module 4: Conference Room Booking System

- Separate booking flow for the 40-seater conference room
- Hourly and daily booking options
- Online calendar showing availability in real time
- Payment processing
- Add-ons: projector, catering, breakout rooms (if available)
- Automated confirmation and access instructions
- Can be marketed separately from the apartments

Module 5: AI Operations Manager

Cleaning automation:

- Guest checks out → system automatically flags apartment for cleaning
- Assigns cleaning staff based on availability
- Staff marks apartment as “ready” → apartment becomes bookable again
- Track cleaning time, frequency, and cost per apartment

Procurement:

- Track consumables per apartment (toiletries, linens, cleaning supplies)
- Auto-generate restock alerts when inventory drops below threshold
- Monthly procurement reports — cost per apartment, cost trends

Staff optimization:

- AI suggests optimal staffing levels based on occupancy patterns
- Low occupancy week? Reduce cleaning staff schedule. High occupancy weekend? Alert for additional support.
- Track staff performance (cleaning time, guest ratings)

Cost analytics:

- Utility costs per apartment (electricity, water — if metered)
- Revenue vs. cost per apartment — which units are most/least profitable
- Monthly P&L per apartment and for the facility overall

Module 6: Facility Management Dashboard

For Court 474 management:

- Bird's eye view of the entire property — 29 apartments + conference room
- Real-time occupancy map (visual grid: occupied, available, cleaning, blocked)
- Financial overview — total revenue, management fees earned, pending settlements
- Maintenance tracker — open issues, resolved issues, cost
- Service charge tracking — which owners have paid, who's overdue

4. TECHNICAL APPROACH

Component	Technology
Frontend	Next.js + React
Backend	Node.js (NestJS)
Database	PostgreSQL
AI Engine	Custom allocation algorithm (fairness scoring, occupancy balancing)
Payments	Paystack (guest payments, owner settlements)
Smart Access	IoT smart locks (auto-generated codes, auto-expiry)
Cloud	Microsoft Azure
Infrastructure	Terraform
Notifications	Email + SMS (bookings, owner reports, staff alerts)
Security	OWASP standard, role-based access control

Algolog has built and deployed property management systems for hospitality businesses across Nigeria — handling bookings, payments, smart lock automation, and operational dashboards. Court474 is a custom build designed specifically for the multi-owner short-let model, with AI-powered guest distribution as its core differentiator.

5. WHAT COURT 474 GETS

Without the system	With the system
Manual guest allocation (WhatsApp, calls, gut feel)	AI distributes guests fairly across all apartments automatically
Owners don't know if they're getting fair share	Every owner sees their stats vs. average in real time
Revenue splitting done manually in Excel	Automatic calculation, settlement reports, audit trail
Front desk needed for every check-in	Smart lock codes sent to guests automatically
Cleaning scheduled via phone calls	Auto-triggered cleaning workflow on checkout
No visibility into costs per apartment	Full P&L; per apartment, per month
Conference room booked via calls/WhatsApp	Online booking with real-time availability and payment
Procurement tracked manually or not at all	Auto restock alerts, usage tracking, cost reports

High staff costs	AI suggests optimal staffing based on occupancy

6. SUMMARY

Court 474's multi-owner model creates a unique challenge: **fairness**. Every owner needs to trust that their apartment is getting its fair share of bookings and revenue. At the same time, Court 474 needs to run operations as lean as possible.

This solution uses AI to solve both problems simultaneously — fair distribution builds owner trust and retention, while automation reduces the staff and manual effort needed to manage 29 apartments and a conference room.

The platform is purpose-built for Court 474's multi-owner model, developed by Algolog — a team with proven experience building and deploying property management systems in the Nigerian market.

WHY ALGOLOG

Algolog Limited is a Nigerian technology company committed to ensuring businesses across Africa get the technology they need to build their future, improve their sales, and strengthen their operations.

- **Spacer** (spacer.so) — Smart hospitality and workspace management platform
- **DeepThread** (deepthread.ai) — AI-powered cross-platform context engine
- **Visora** (visora-ai.com) — AI-powered data analytics and insights
- **PrivateTransfer** (privatetransfer.ng) — Secure transfer platform

NEXT STEP

We'd love to have a conversation about how Court474 can work for your specific operation — the apartment mix, the owner agreements, the conference room, and how AI distribution would integrate into your day-to-day workflow.

Whenever you're ready, reach out to:

- **Call/WhatsApp:** Fayinminu Femi — +234 705 301 6348
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